USER INTERFACES FOR INFORMATION ACCESS

Human-Computer Interaction · The Information Access Process · Starting Points
Query Specification · Context · Relevance Judgments · Interface Support for Search

What Makes an Interface Effective?

"When an interactive system is well designed, the interface almost disappears, enabling users to concentrate on their work, exploration, or pleasure." *Ben Shneiderman*

Informative feedback

show how query and results relate

Easy reversal

undo, go back, try again

Internal locus of control

user decides when and how

Reduce memory load

keep track of choices made

Novice and expert modes

simple first, power later

Real life: Google's single empty box for beginners, with Advanced Search one click away for experts, is the 'scaffolding' idea in practice.

The Role of Visualization

- Humans are highly attuned to images: a photograph says more about a face than a written description, and a scatter plot says more than a table of numbers.
- *Hard case: visualizing text, because language carries abstract ideas with no physical shape.*

Brushing and linking: two views of the same data; select in one, highlights in the other

Panning and zooming: move across a scene, or move in and out for detail

Focus plus context: enlarge what you look at, shrink the rest, like a fisheye lens

Magic lenses: transparent windows that transform what lies beneath

Overview plus details: table of contents in one pane, chapter text in another

Real life: Google Maps zoom (panning/zooming), the macOS Dock magnifying icons (focus plus context), Gmail's folder list beside the message (overview plus details).

Evaluating Interactive Systems

Precision and recall
(built for non-interactive ranking, TREC-style top 1000)

Interactive systems: users need a few good documents, not high recall

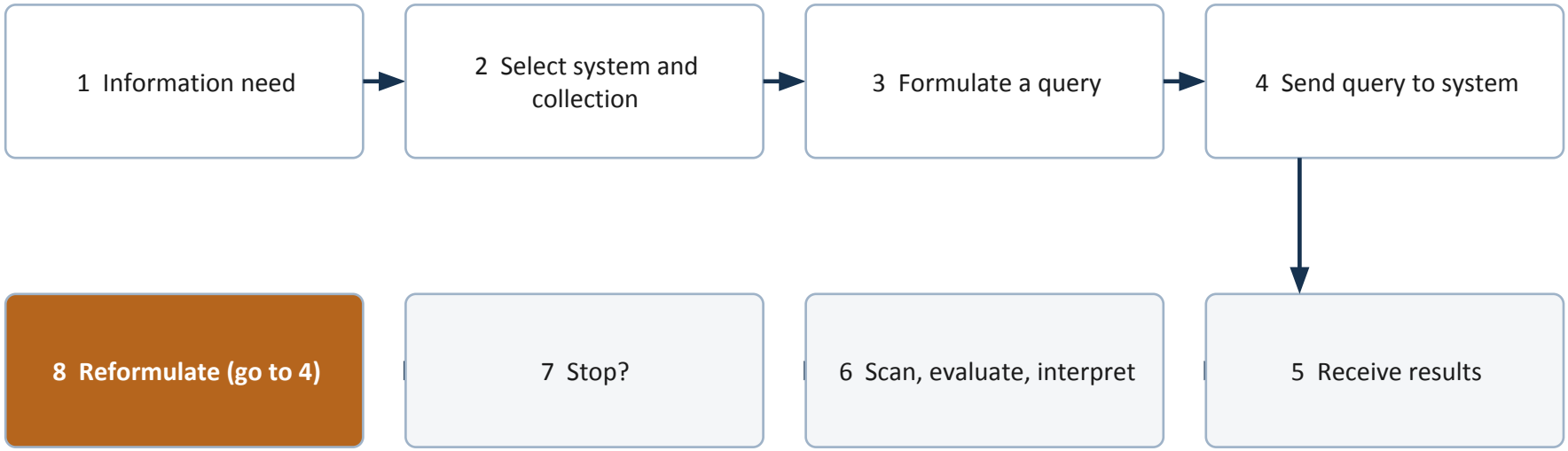
so measure instead:

- Time to learn the system
- Time to complete benchmark tasks
- Error rate
- Retention of use over time

- Heuristic evaluation (Nielsen): judge the interface against general properties, without chasing statistically significant results.

Real life: A/B tests at Google or Amazon track task completion time and drop-off, not just 'was the result relevant'.

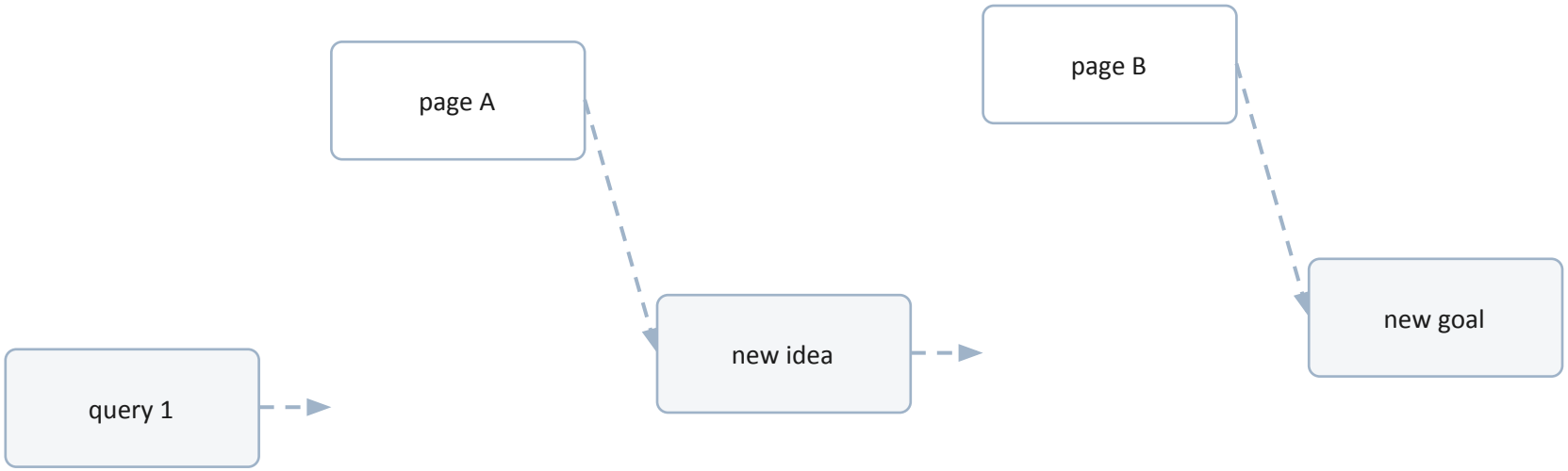
The Standard Model of Information Access



- This loop is exactly what a Web search engine shows you: type, scan the list, retype.
- **Hidden assumptions: the need never changes, and the goal is to refine until only relevant documents remain.**

Real life: searching 'best laptop', scanning, then refining to 'best laptop under 60000 for coding'.

Berry-Picking: How People Really Search



- **Bates' model: needs and queries keep shifting as you read; value comes from bits gathered along the way, not one final result set.**
- O'Day and Jeffries: searches form a connected series on one theme; each result can trigger a new goal; the learning matters more than the last result list.
- Interface implication: let users record progress, save intermediate results, and follow unexpected trails.

Real life: researching a phone, you open reviews, discover you care about battery, then switch to comparing chargers.

Scan, Select, Query, Navigate, Browse

Activity	Meaning in this chapter	Real-life example
Scanning	looking over titles, categories, links, clusters	skimming a results page
Querying	producing a new, ad hoc set of information	typing into the search box
Selection	picking already-organised information	clicking a category
Navigation	following a chain of links toward a goal	clicking through Wikipedia links
Browsing	casual, undirected exploration	scrolling a YouTube home page

- Search is only part of the work: O'Day and Jeffries found 80% of analysis fell into six types: finding trends, comparing, aggregating, identifying a critical subset, assessing, interpreting.
- So the whole process has two halves: search/retrieval and analysis/synthesis.

Why Older Interface Studies Don't Fully Apply

Earlier studies	Modern conditions
Bibliographic citations only	Full text available
Boolean systems	Statistical ranking
Professional intermediaries	End users, often new to searching
Separate, costly-to-switch collections	Tens of thousands of linked sources
TTY command lines	Bit-mapped, direct manipulation

A near miss is far more useful on the Web, because a hyperlink can carry you from the near miss to what you needed.

Starting Points: Lists and Overviews

- Users with a vague need often cannot write a query yet; they need somewhere to begin.
- **Lists of collections: a menu of available sources to choose from.**
- **Overviews: a picture of what a collection contains, before searching it.**

Category / directory overview	Automatically derived overview	Co-citation clustering
Arts > Music > Rock (Yahoo! directory)	clusters of similar documents (Scatter/Gather)	group papers that are cited together

Real life: Amazon's department menu, Google Scholar's list of sources, a library's 'databases A to Z' page.

Wizards and Source Selection

Examples, dialogs and wizards

sample queries, step-by-step questions that build a query for you

Automated source selection

the system decides which collections are worth searching

- Kohonen feature-map overview study: users liked jumping between areas without backing up (unlike a directory), but some wanted a visible hierarchy, zoom, or alphabetical order.
- Takeaway: no single overview suits everyone, so offer more than one.

Real life: Google's suggested searches and 'People also ask'; metasearch tools that query several engines at once.

Why Boolean Queries Confuse People

"dogs AND cats" everyday meaning: more things	**logical meaning: documents about BOTH, so fewer results**
"tea OR coffee" everyday meaning: pick one	**logical meaning: either, or both**

- Parentheses and operator precedence are unfamiliar to most users.
- Web fixes: 'all the words' / 'any of the words' choices, and the AltaVista '+' prefix (often misread as an infix AND).
- Pure Boolean also does no ranking: a document either matches or does not.

Real life: Google quietly treats a multi-word query as AND; the '-' sign is its simple NOT.

From Command Lines to Forms; Faceted Queries

```
FIND PA darwin AND TW species
```

command language: remember every command and attribute

```
 Author  [          ]
Title     [          ]
  Year      [  v  ]
```

form fill-in and menus: options visible

Faceted query: one topic per line, OR inside a line, AND across lines

```
(osteoporosis OR "bone loss")
```

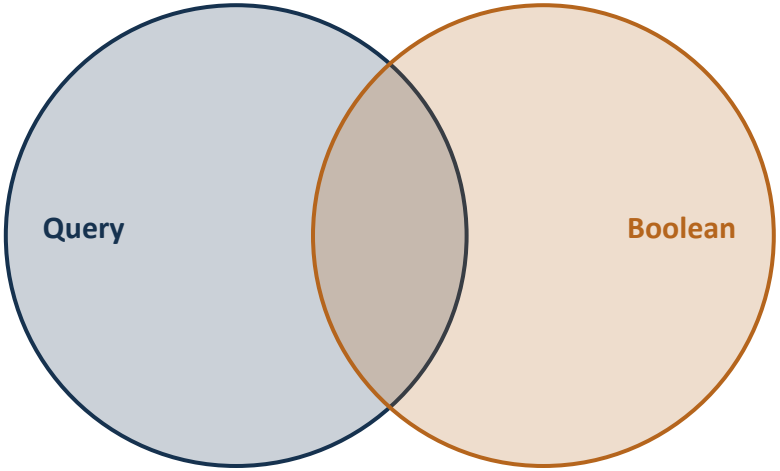
```
(drugs OR pharmaceuticals)
```

```
(prevention OR cure)
```

- Quorum ranking: rank documents by how many facets they cover, so one mention of each of three facets beats thirty mentions of one.

Real life: Google Advanced Search, library catalogue forms, and e-commerce filters are form-based, faceted querying.

Graphical Query Specification



overlap = AND
Circle only = OR
Unselected oval = NOT

VQuery: ovals for terms; where you click decides the operator.

- Direct manipulation: continuous view of the object, clicks instead of syntax, quick reversible actions with visible effect. Studies found Venn-style interfaces faster and more accurate than command syntax.
- Other designs: filter-flow (water-width shows matches), Anick's blocks (rows AND, columns OR), magic lenses (stack lens = AND).
- **Query preview: show results of variations cheaply, e.g. words matching the letters typed so far.**

Real life: search autocomplete is a query preview, updating with every key you press.

Phrases, Proximity and Natural Language

`white w/3 house` `(LEXIS-NEXIS)`	`"white house"` `(quotation marks)`	president Lincoln matches 'President Abraham Lincoln'

- Two words as an exact phrase can turn a mess of results into a short, relevant list.
- Best idea: treat a phrase as a small proximity range, with weight falling off as the words move apart.

Natural language and free text

- Ranking lets users type naturally, but gives little feedback on why a document ranked where it did.
- Question answering: Murax finds sentences holding the question's noun phrases (answer: Norman Mailer); FAQ Finder matches question-answer pairs; Ask Jeeves matched hand-built question templates.

Real life: Google's direct answer boxes and voice assistants are the descendants of Murax and Ask Jeeves.

Context: Surrogates, Highlighting and KWIC

Information Retrieval - Wikipedia

en.wikipedia.org · 14 Mar 2025 · score 0.91

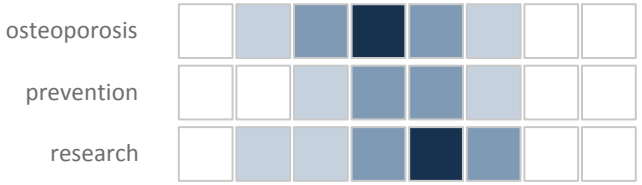
...is the process of obtaining **information** system resources relevant to an **information** need... A **retrieval** query is a formal statement...

- Document surrogate: title plus metadata (date, source, score) and a short excerpt.
- Highlighting query words is repeatedly shown to help; KWIC (keyword in context) shows the sentences containing them, which differs from an abstract.
- Best fragments: near the start and containing most query terms, shown in original order.

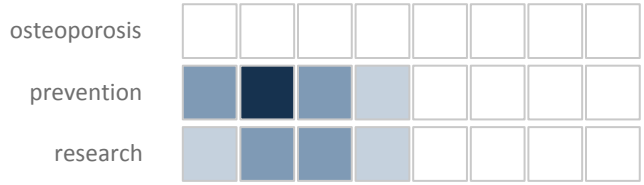
Real life: every Google result shows bold query words inside a snippet. Manning Ch.8 section 8.7 (results snippets).

Showing Where Query Terms Hit

TileBars: each row is a facet, each column a 100-word passage, darker = more hits



overlap in the middle: likely relevant



never mentions osteoporosis: skip

- **Between documents:**
  - InfoCrystal shows how many documents contain each subset of terms (hard beyond four terms); VIBE and Lyberworld place icons between term anchors.
- **SeeSoft**
  - draws text as coloured line strips, originally for source-code changes.

Real life: GitHub's blame view and VS Code's minimap borrow the SeeSoft idea; Ctrl+F highlights every match with a tick on the scrollbar.

Context from Structure, Categories and Links

System	Context it provides	Idea
SuperBook	Hits shown inside the table of contents	focus plus context
CORE / DynaCat	Results grouped under category (MeSH) hierarchy	category context
SONIA	Clusters, adjusted by the user, then train a classifier	unsupervised + supervised
Cha-Cha	Shortest link paths from the site root to each hit	organise an intranet
Mapuccino	Node-and-link map of the crawled site	link structure view
Envision / Table Lens	Results in a table by metadata (author, date)	tabular context

Real life: Gmail's search result with labels, PubMed results filtered by MeSH terms, a website's breadcrumb trail.

Interfaces for Relevance Feedback

```
[x]  Paper A  (marked relevant)
     [ ]  Paper B
[x]  Paper C  (marked relevant)
     [ ]  Paper D

   [ Find more like these ]
```

- Original form: mark relevant documents; the system revises the query and re-ranks.
- Web simplification: a one-click 'more like this'.
- Aalbersberg's incremental feedback works with a single document at a time.

Koenemann and Belkin: how much should users see 'under the hood'? (64 subjects, average feedback cycles)



Penetrable (user picks which suggested terms to add) was best. Real life: Amazon 'customers also viewed', Pinterest 'more like this'.

Beyond Explicit Feedback

Idea	How it works	Real-life echo
Background assistants	Watch behaviour, predict next need (Letizia, Syskill & Webert, Butterfly)	browser link suggestions, citation tools
Implicit feedback	A bookmark counts as positive; skipped links as negative	YouTube watch time
Group judgments	Use ratings of people with similar tastes, ignore content	Netflix, Spotify recommendations
Pseudo-relevance feedback	Assume the top-ranked documents are relevant and expand the query	automatic query expansion

Pseudo-feedback works best when the original query is long and precise. Manning Ch.9, section 9.1.6.

String Matching and Window Layout

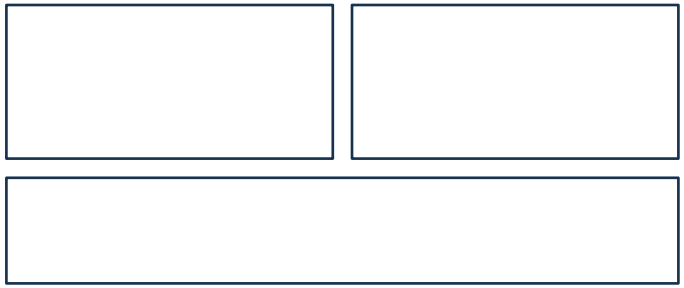
```
Find:  retrieval   [^] [v]
```

- Simple 'find': no ranking, no Boolean; stale results while typing a new query is a design trap (best fix: stack old searches).
- Help early: spell-check low-frequency terms, suggest thesaurus terms as the user types.



Monolithic: fixed panes

- Overlapping windows: flexible but quickly crowded. Workspaces group related windows per task and persist across days; Elastic windows resize whole groups.



Tiled windows

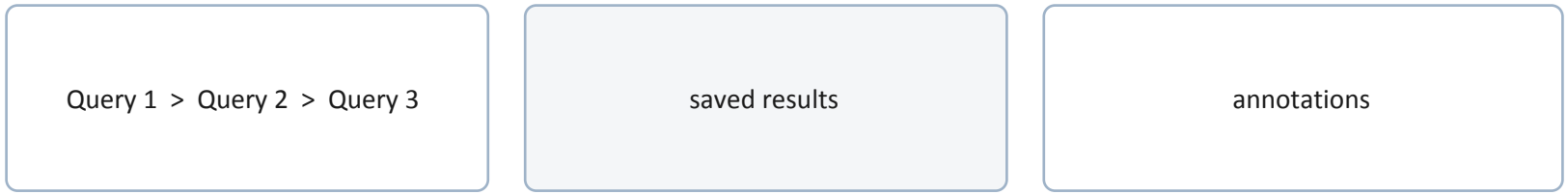
Real life: Ctrl+F in a browser; macOS Spaces and Windows virtual desktops are workspaces.

Four Example Interface Designs

System	Layout idea	Memorable detail
InfoGrid	Monolithic: query forms, results centre, document viewer right, history below	holding area to keep documents
SuperBook	Table of contents left, text right, hit counts per outline level	redesigned after reading user logs
DLITE	Drag a query icon onto a source icon; results as a circular pool	icon shakes 'no' on an illegal drop
SketchTrieve	A canvas; arrange results side by side like a spreadsheet	keeps half-finished ideas visible

Real life: Figma and Miro canvases (SketchTrieve), Slack's channel-plus-thread layout (monolithic panes).

Search History and Integrating the Pieces



- Show available moves and past moves; let users annotate, bundle and reload sessions. A hierarchical Web history needed fewer page visits to get back to a page.
- Scanning, selection and querying should flow into each other: output of one search should feed the next, even across sessions.
- Hertzum and Frokjaer: users like mixing scanning and querying, but unguided mixing was wasteful, so offer flexibility inside constraints.
- **Cat-a-Cone: search and browse a huge category tree together; the Radiation example showed why the term sat under Diseases (occupational hazards).**

Real life: browser history and 'search within results'; Google Scholar's saved library; Melvyl's 'revise earlier query' checkbox.